

Taylor tail control for the phase factor

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Abstract

Unit 199, the first unit of programme A (general- d phase-dressed leading term). Rather than the rescaled-measure route proposed in Astra #20, the programme expands the phase factor $e^{\beta s \xi(u)}$, $s = Nu^k$, in a Taylor series so that every term is the existing continuous-amplitude theorem at shifted exponents $h + jk$; Astra #21 endorsed this (“GO with Taylor-in-phase”) and corrected the assembly (bound $|C - P_K|$ by the same tail estimate against the limiting measure; avoid `tsum` in the headline). This unit provides the elementary tail estimates: the Taylor remainder $|e^x - \sum_{j < J} x^j / j!| \leq |x|^J e^{|x|} / J!$ for all real x ; $s^{2K} e^{-cs^2} \leq K! / c^K$; completing the square $e^{bs - \beta s^2} \leq e^{b^2/(2\beta)} e^{-\beta s^2/2}$; the tail constant $\tau_K = (b^2)^K / (2K)! \cdot K! / (\beta/4)^K \cdot e^{b^2/(2\beta)} \rightarrow 0$; and the uniform bound $|e^{sx} - \sum_{j < 2K} (sx)^j / j!| e^{-\beta s^2} \leq \tau_K e^{-\beta s^2/4}$ for $s \geq 0$, $|x| \leq b$. Zero sorry/axiom; 9076 jobs.

1 The things formalised

```
theorem abs_exp_sub_sum_le (x : ℝ) (J : ℕ) :
  |exp x - ∑ j in range J, x^j / j!| ≤ |x|^J / J! * exp |x|
theorem pow_mul_exp_neg_sq_le (hc : 0 < c) (K : ℕ) : s^(2K) * exp (-(c s^2)) ≤ K! / c^K
theorem exp_linear_sub_sq_le (h : 0 < β) :
  exp (b s - β s^2) ≤ exp (b^2/(2β)) * exp (-(β/2 s^2))
noncomputable def tailCoeff (b : ℝ) (K : ℕ) : ℝ :=
  (b^2)^(K/2) / (2K)! * (K! / (β/4)^K) * exp (b^2/(2β))
theorem tendsto_tailCoeff (h : 0 < β) : Tendsto (fun K => tailCoeff b K) atTop (0)
theorem phase_tail_le (h : 0 < β) (hs : 0 ≤ s) (hx : |x| ≤ b) (K : ℕ) :
  |exp (s x) - ∑ j in range (2K), (s x)^j / j!| * exp (-(β s^2))
  ≤ tailCoeff b K * exp (-(β/4 s^2))
```

2 Mathematics

The remainder bound is the crude series estimate $\sum_{i \geq J} |x|^i / i! \leq (|x|^J / J!) \sum_i |x|^i / i!$ from $(i + J)! \geq i! J!$; Mathlib’s own `Real.exp_bound` needs $|x| \leq 1$, which is useless here because s is unbounded. The even truncation order $2K$ is chosen so that the polynomial factor s^{2K} can be absorbed by half of a Gaussian, $s^{2K} e^{-cs^2} \leq K! / c^K$, without any Gamma-function asymptotics; the price $(2K)! \geq (K!)^2$ then makes $\tau_K \leq (4b^2/\beta)^K / K! \cdot e^{b^2/(2\beta)}$, and `FloorSemiring.tendsto_pow_div_factorial_atTop` finishes.

3 Lean notes

- `NormedSpace.expSeries_div_hasSum_exp x` has the algebra implicit; passing `α` explicitly makes Lean read `α` as the algebra element. `Real.exp_eq_exp_` bridges to `Real.exp`.

- The `tail-of-series` lemma is `Summable.sum_add_tsum_nat_add` (dot notation); `sum_add_tsum_nat_add` is only mentioned in a docstring.
- `set c := / 4` did not fold the occurrence produced by `unfold tailCoeff` (different elaboration of the literal), so `div_pow` hit the wrong power. Use `conv_rhs => rw [div_pow, div_div]` to scope the rewrite instead of folding.
- `gcongr` closes the side goal `0 < |x|` itself; an explicit bullet afterwards errors with “no goals”.

4 Next

Unit 200: shifted-exponent transport ($J, m, \text{faceProj}, \text{residualWeight}$ invariant under $h \mapsto h + jk, \lambda \mapsto \lambda + j/2$; the term theorem with the N^j and 2^{m-1} bookkeeping). Unit 201: the phase-dressed normal moment $S_{p/2}^{(\beta)}(a) = \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta s a} ds$ with its Gaussian moments and finite Taylor approximation. Unit 202: the leading-term theorem by direct approximation to the explicit coefficient.